NLP for Structured HAZOP Reasoning

Research Report

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Abstract

Hazard and Operability Study (HAZOP) reports encode process-safety knowledge in a structured but predominantly tabular form. This report examines whether natural-language processing and large language models can support reasoning over that knowledge rather than only extracting labels from report text. A survey of HAZOP automation literature is combined with a detailed analysis of an ontology-based HAZOP reasoner. The review shows that causal, multi-step, branching inference is already possible when process and safety knowledge are manually encoded. The remaining research question is therefore narrower: whether causal knowledge can be reconstructed from unstructured historical HAZOP reports and then constrained or validated using structured reasoning. This distinction provides a more defensible foundation for future system design and evaluation.

Keywords: HAZOP; process safety; natural-language processing; knowledge graphs; ontology reasoning; causal inference; large language models

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Week 1 Background, Literature Survey, and Research Framing

This section establishes HAZOP fundamentals, defines the project framing, and surveys prior work on automated HAZOP analysis. The review shows that the research problem must be stated more narrowly than the initial framing.

1.1 Project Motivation

HAZOP (Hazard and Operability Study) is a structured method for identifying hazards and operability issues in chemical processes. Reports are typically authored manually and stored as tabular worksheets. This project investigates whether NLP/LLM methods can support structured reasoning over the knowledge contained in HAZOP reports, rather than treating HAZOP text purely as a source for information extraction.

1.2 Fundamentals of HAZOP

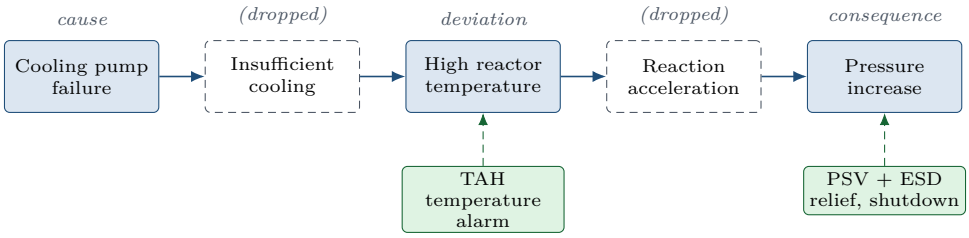
- **Node**: a defined section of a process (e.g., a vessel, a line between two points) selected for study.
- **Parameter**: a process variable examined at a node (e.g., flow, temperature, pressure, level).
- **Guide word**: a standardized qualifier (e.g., NO, MORE, LESS, REVERSE) combined with a parameter to systematically generate deviations.
- **Deviation**: a guide word applied to a parameter (e.g., "MORE temperature").
- **Cause**: a condition or event that could produce the deviation.
- **Consequence**: the potential outcome if the deviation occurs and is not mitigated.
- **Safeguard**: an existing protection (alarm, interlock, relief device, procedure) that reduces likelihood or severity.

1.3 HAZOP as a Reasoning Structure

A HAZOP worksheet presents Cause, Deviation, Consequence, and Safeguard as flat table columns. This is a presentational simplification: the underlying process-safety knowledge can involve intermediate events, multiple contributing causes, branching consequences, and multiple safeguards acting at different points in a chain. Figure 1.1 contrasts the two views for a single worksheet row.

Deviation	Cause	Consequence	Safeguards
MORE temperature	Cooling pump failure	Reactor overpressure	TAH; PSV; ESD

(a) the worksheet row: four flat fields, no ordering between them



(b) the underlying chain: dashed boxes are intermediate events the worksheet has no column for; safeguards attach to specific events

Figure 1.1: The same hazard scenario as a flat worksheet row (a) and as a causal chain with event-specific safeguards (b). The flattening loses the intermediate events and the attachment point of each safeguard — which is what makes the row hard to reason over automatically.

Representing this as a graph rather than a table is *not* itself a novel observation — it is a modeling choice that recurs across the digraph, ontology, and knowledge-graph literature (Categories A, B, D below). What varies across papers is not whether a graph representation is used, but how the graph is populated (manual encoding vs. text extraction) and what operations are performed on it (traversal vs. learned reasoning vs. free generation).

1.4 Initial Project Objective (Provisional)

Initially hypothesized objective: extract causal structure from unstructured historical HAZOP report text and represent it as a graph capable of supporting inference of missing causes, consequences, or safeguards. This

objective is recorded here for continuity only — it predates the literature findings below and should not be treated as validated or final.

1.5 Literature Survey

The initial literature set was organized into five provisional categories. Figure 1.2 shows what distinguishes them in practice: where each family gets its structure from, and what it then does with it.

A. Classical model-based / expert-system HAZOP. Vaidhyanathan & Venkatasubramanian [14, 15]; Rahman et al., ExpHAZOP+ [10]; Rodríguez & de la Mata, D-higraphs [11]; Taylor [13]. These operate on hand-built process models rather than natural-language text; reasoning proceeds via digraph traversal or rule-chaining over a manually constructed representation.

B. Ontology-based causal reasoning. Aziz et al. [1]; Single et al. [12]. Causal relationships between HAZOP concepts are explicitly encoded in an ontology and processed by a semantic reasoner.

C. NLP/ML applied to HAZOP text. Feng et al. (2021, NLP in HAZOP reports) [5]; Wang et al. (2021, active learning/text mining) [17]; Wang & Gu (2022, data mining) [16]; Ekramipooya et al. [2, 3]. These extract or classify information from HAZOP text (entity recognition, deviation-category prediction, recommendation clustering) without constructing or reasoning over an explicit causal graph.

D. Knowledge graphs and structured HAZOP reasoning. Mao et al. [9]; Feng et al. (2021, causal ontology description model) [4]; Wang et al. (2022, ISKG/HAINEX) [18]; Li & Zhao (2024) [7]; Li & Zhao (2025, HAZOPCT) [8]. These construct HAZOP knowledge graphs, in several cases via NLP-based entity extraction from text, and apply graph-based reasoning methods over the resulting structure.

E. LLM-based HAZOP generation. Lee et al. [6]. Multimodal LLMs generate complete HAZOP worksheets end-to-end from a P&ID and a standardized prompt, without an intermediate structured representation.

These categories are provisional and may be revised as the survey deepens.

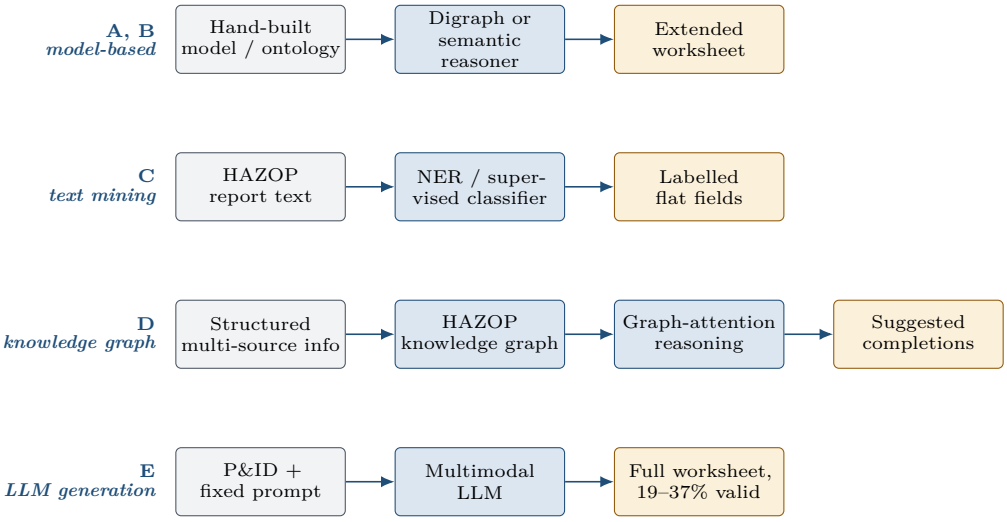


Figure 1.2: The five categories differ less in whether they use a graph than in where the structure comes from and what is done with it. Grey boxes are inputs, blue boxes are processing steps, amber boxes are outputs. Only Category C starts from report text, and it is the one family that never builds a graph.

1.6 Capability Analysis

The most important comparison axis identified in this review is the distinction between five operations that are frequently conflated in this literature:

1. **Extraction** — does the method pull information out of text (entities, spans, labels)?
2. **Representation** — does it explicitly represent relationships (graph, ontology), or only flat fields?
3. **Causal reasoning** — does it operate over those relationships to derive new statements, or merely store them?
4. **Inference** — does it produce previously unstated information (a missing cause, consequence, or safeguard)?
5. **Validation** — is the inferred output checked against expert judgment, process knowledge, or another ground truth?

A paper should not be classified as performing causal reasoning merely because its output table contains fields labeled “cause” and “consequence.” Applying this test across the five categories gives Table 1.1.

Table 1.1: Capability matrix. ✓ = performed, ∼ = partial or method-dependent, × = not performed. *Validation* records whether validation was attempted, not whether it succeeded — Category E validates carefully and reports a poor result. The last row is the profile this project would have to occupy to be distinct; it is a target, not a result.

Cat.	Primary input	Extr.	Repr.	Causal reas.	Infer.	Valid.
A	Hand-built process model (digraph, D-higraph)	×	✓	✓	✓	∼
B	Manually authored ontology	×	✓	✓	✓	✓
C	HAZOP report text	✓	×	×	×	∼
D	Structured multi-source information + ontology	∼	✓	✓	✓	✓
E	P&ID + standardized prompt	n/a	×	∼	✓	✓
Target	Unstructured historical report text	✓	✓	✓	✓	✓

Under this test, the Category C papers (Ekramipooya et al. 2023, 2024; Wang & Gu 2022) perform extraction and classification but not causal reasoning: for example, Ekramipooya et al. (2023) predicts a deviation *category* from cause text via supervised classification on a small dataset (three process nodes, 25 records, 42 after oversampling), which is correlational pattern-matching rather than reasoning over a causal structure.

1.7 Important Findings

1.7.1 Single et al. (2020)

This paper already provides ontology-based automated HAZOP reasoning. It models semantic/causal relationships between HAZOP concepts and uses a semantic reasoner to generate extended HAZOP worksheets, explicitly addressing the risk of automatically generated but meaningless scenarios through semantic constraints. Validation was performed on a hexane storage tank case study, where the quality and quantity of the automatically generated results agreed with the original expert worksheets.

Constraint on the write-up

The report must not claim that existing work fails to perform causal HAZOP reasoning. Single et al. (2020) already does, and validates it against expert worksheets.

Open questions requiring deeper reading: What exactly is the input (free text vs. pre-encoded knowledge)? How much of the ontology is manually authored per new process? How are causal chains represented internally? What precisely does the inference mechanism infer, and how dependent is it on predefined process knowledge?

1.7.2 HAZOPCT — Li & Zhao (2025)

This paper is closer still to the initially proposed direction. It constructs a HAZOP knowledge graph from multi-source process-safety information organized under a predefined ontology, then applies graph-attention-based reasoning over that graph to generate recommended conclusions for completing HAZOP analysis — i.e., it performs missing-link inference, not just storage. It is evaluated on a sulfur-recovery-process case study, with ablation studies validating the contribution of semantic information and the graph-attention mechanism, and attention weights used to provide interpretability of the reasoning process.

Constraint on the write-up

The report must not claim that knowledge-graph-based HAZOP inference is unexplored. HAZOPCT performs exactly this, with ablations and an interpretability story.

Open questions requiring deeper reading: What is the raw input, and how much of the graph is manually curated versus text-extracted? What exactly are the node and edge types? What does "completeness" mean operationally in this paper? Does it reconstruct causal chains directly from unstructured HAZOP report text, or from already-structured data? Does it explicitly model intermediate causal events, or only direct cause–consequence links?

1.8 Revision of Initial Assumptions

The initial framing — "existing work only extracts HAZOP information, while this project introduces causal reasoning" — is not supportable given the findings above and has been discarded. The literature survey instead narrows the research problem before system design and evaluation.

1.9 Provisional Research Direction

One possible direction that survives the literature review, stated here only as a hypothesis, is reconstructing causal chains from *unstructured, historical* HAZOP report text (as opposed to pre-encoded ontological knowledge)

using LLM-based extraction, with the resulting chains constrained or validated against a structured graph rather than accepted at face value. Figure 1.3 sketches the shape of this pipeline and marks the two points where it would differ from existing work.

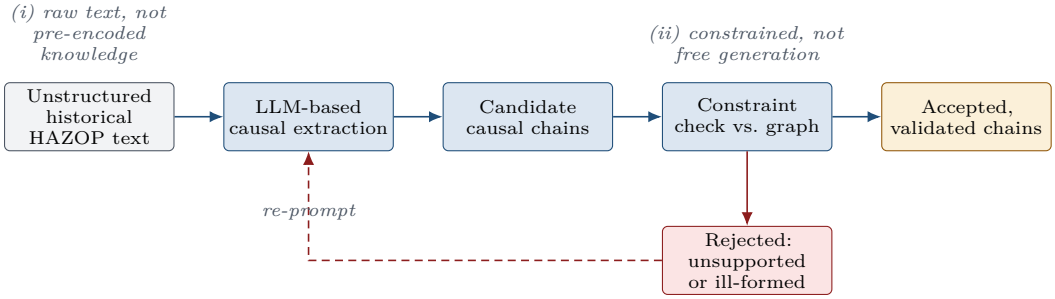


Figure 1.3: Provisional pipeline. The two annotations are the only points at which it currently claims to differ from existing work, and both depend on readings that have not yet been done: (i) assumes Single et al. and HAZOPCT work from pre-structured knowledge, and (ii) responds to the low scenario validity of unconstrained LLM output.

This is motivated by two observations that still require confirmation: (i) Single et al. and HAZOPCT appear to reason over knowledge that is substantially pre-structured rather than reconstructed from raw report text; (ii) Lee et al. (2026) [6] found that unconstrained LLM generation of full HAZOP worksheets achieved high textual similarity to expert references but only 19–37% scenario validity, with safeguards skewed toward procedural rather than design-level interventions, suggesting raw LLM output is not reliable without some form of structural constraint.

Status of this direction

This is not yet a defensible novelty claim and must not be presented as the final research gap until Single et al. (2020) and HAZOPCT have been read in full. Both premises (i) and (ii) above are assumptions about papers not yet read closely.

1.10 Open Questions

- Does Single et al. (2020) operate on free text at all, or exclusively on pre-encoded ontological knowledge?
- What is the actual node/edge schema in HAZOPCT, and does it model intermediate causal events or only direct links?
- Is "completeness" in HAZOPCT defined relative to a fixed schema of expected fields, or relative to open-ended causal chain reconstruction?
- How much manual encoding effort does each approach require per new process, and does this scale to arbitrary historical report corpora?
- Does either paper attempt cross-report generalization, or are results confined to a single case-study process?

1.11 Directions for Further Research

1. Full read of Single et al. (2020), answering the open questions listed above.
2. Full read of Li & Zhao (2025, HAZOPCT), answering the open questions listed above.
3. Re-evaluate the provisional research direction in light of these two full reads.
4. Only after this: attempt a first defensible statement of the research gap.

Week 2 Ontology-Based HAZOP Reasoning: A Detailed Analysis

2.1 Objective

The objective was to understand Single et al. (2020) in enough depth to state precisely what the system does, what it assumes, and what it does not attempt. The analysis distinguishes throughout between what the paper itself establishes and our interpretation of its relevance to this project.

2.2 Paper Studied

Single, J. I., Schmidt, J., and Denecke, J. (2020). "Ontology-based computer aid for the automation of HAZOP studies." *Journal of Loss Prevention in the Process Industries*, 68, 104321 [12].

2.3 Problem Addressed by Single et al.

The paper addresses the automated *generation* of HAZOP worksheets for a given process unit, using a formally represented ontology of HAZOP-relevant knowledge together with a symbolic inference engine. As established in the literature survey, this is a generation problem, not an extraction problem: the system is not given a historical HAZOP report and asked to recover its reasoning; it is given a process-unit model and an ontology of causal/safety knowledge, and asked to produce a HAZOP worksheet from scratch.

2.4 System Architecture

The system requires, as input, structured knowledge about a process unit and its context: the process unit itself, its equipment, the substance(s) involved and their relevant properties, the process, and the circumstances under which it operates. **This knowledge is already available to the inference system before reasoning begins** — the system does not discover it from raw text. Figure 2.1 summarizes the architecture at this level.

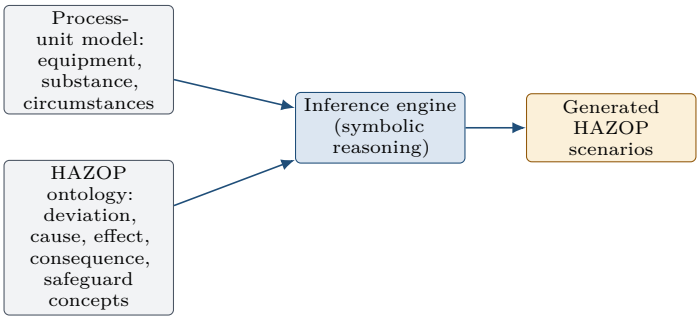


Figure 2.1: The process-unit model and the HAZOP ontology are both supplied as structured input; the inference engine consumes both to produce scenarios. Neither the process-unit model nor the ontology is derived from unstructured report text in this system.

Our interpretation: this places the entire knowledge-acquisition burden — building the process-unit model and populating the ontology — outside the automated part of the system. What is automated is the reasoning step, not the knowledge-entry step.

2.5 HAZOP Ontology

The ontology represents the concepts needed for HAZOP reasoning: Deviation, Cause, Effect, Super Cause, Consequence, Safeguard, Process Unit, Equipment, Substance, Process, and Circumstances. Critically, these concepts are not flat fields — the ontology encodes a causal structure among them, and this structure supports branching and multiple causes, not only single linear chains. Figure 2.2 shows the general shape of the causal structure the ontology supports.

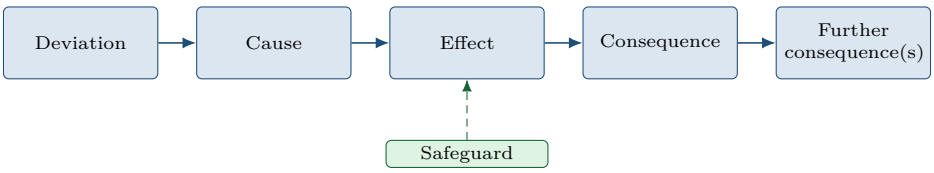


Figure 2.2: The ontology's general causal structure. This is a template for the concepts and their ordering; the actual case-study instantiations (Section 2.8) show branching and multiple-cause variants of this same structure.

Because the ontology encodes this structure explicitly rather than as flat worksheet fields, it is closer to a

structured causal representation than to a conventional HAZOP table — sharpening the observation that a flat worksheet is a presentational simplification of an underlying graph.

2.6 Inference Engine

The system's reasoning is implemented in OWL 2 DL, executed with the HermiT reasoner via Owlready2 in Python. **FACT:** this is symbolic, ontology-based reasoning, not statistical machine learning and not an LLM. The inference sequence described in the paper proceeds through five steps for a given deviation: identify causes, identify effects, identify super causes, identify consequences, and identify safeguards. The reasoner derives these by evaluating logical consequences of the modeled process information against the ontology's axioms, not by pattern-matching against text.

2.7 Causal Reasoning Representation

The paper's causal representation goes beyond simple one-to-one Deviation → Cause or Cause → Consequence links in three ways that are worth recording explicitly, since they matter for comparing this system against later knowledge-graph literature.

Multi-step chains. A single deviation can propagate through several intermediate concepts before reaching a final consequence (Deviation → Cause → Effect → Consequence → further consequence(s)), matching the general structure in Figure 2.2.

Branching. A single effect can lead to more than one distinct consequence path. Figure 2.3 shows this using the loss-of-containment example from the case study (Section 2.8).

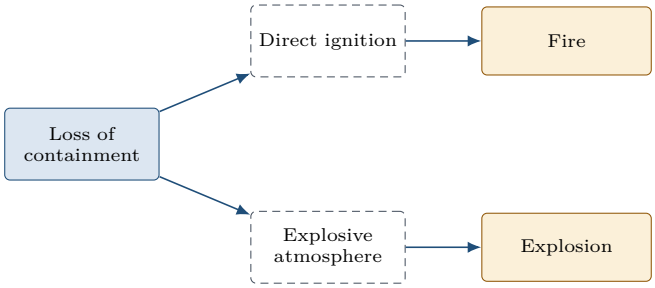


Figure 2.3: A single effect (loss of containment) branching into two distinct consequence paths, as represented in the ontology.

Multiple causes. The ontology also supports several distinct causes converging on a single effect, rather than assuming each effect has exactly one cause.

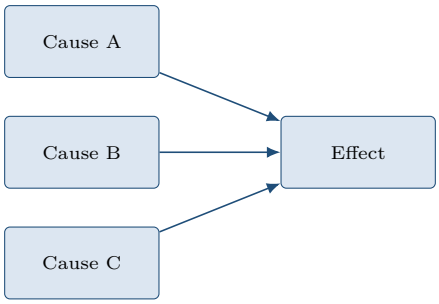


Figure 2.4: Multiple, independently modeled causes converging on a single effect.

Our interpretation: this branching/multi-cause capability is a genuine capability of the ontology's structure, not merely of the reasoning algorithm; it means the representation itself, not just the inference procedure, departs from a flat worksheet.

2.8 Hexane Storage Tank Case Study

The methodology is demonstrated on a hexane storage tank. The process-unit model includes equipment such as valves, drain lines, a level control system, and a transfer/recirculation pump, together with hexane's safety-relevant properties. The original, human-authored HAZOP for this tank considers eight deviations: high level, low level, high temperature, low temperature, high pressure, low pressure, high concentration of contaminants, and loss of containment. The automated system generated 40 potential scenarios across these eight deviations.

One representative generated causal chain, given in the paper for the high-level deviation, is:



The generated worksheet is structured with more fields than a conventional flat HAZOP table: Unit, Substance, Deviation, Super Cause, Cause, Effect, Consequence, and Safeguard, in place of the conventional Deviation / Cause / Consequence / Safeguard columns. Table 2.1 contrasts the two.

Table 2.1: Field structure of a conventional HAZOP worksheet compared with the fields in Single et al.'s generated worksheet. The additional fields (Unit, Substance, Super Cause, Effect) make intermediate causal steps explicit that the conventional table leaves implicit.

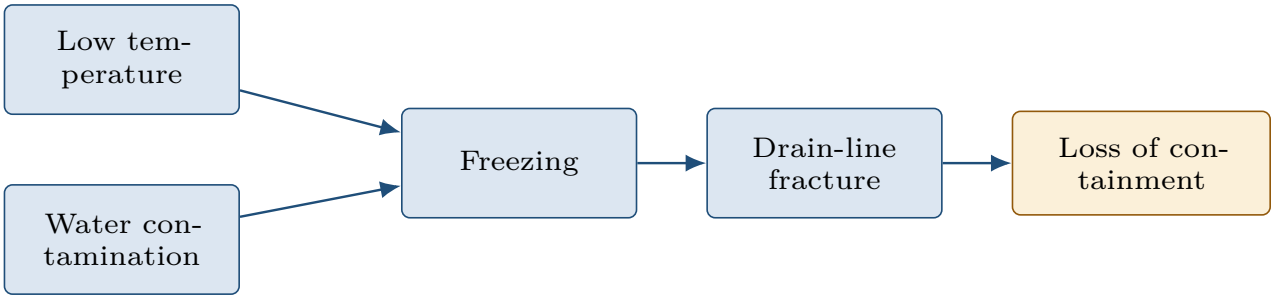
Conventional worksheet	**Single et al. generated worksheet**
Deviation	Unit
Cause	Substance
Consequence	Deviation
Safeguard	Super Cause
	Cause
	Effect
	Consequence
	Safeguard

2.9 Automated HAZOP Results

Beyond the field structure itself, the case study illustrates two representational points worth recording precisely.

Branching example. The loss-of-containment scenario shown in Figure 2.3 (direct ignition → fire; explosive atmosphere → explosion) is drawn directly from the case study's generated results, not a constructed illustration.

Guideword overlap. The paper also discusses scenarios spanning more than one nominal deviation. A freezing-related scenario combines low temperature and water contamination as joint conditions:



Our interpretation: this shows a physical scenario does not always correspond cleanly to one deviation; a representation limited to "one deviation → one scenario" would not capture this case. Whether a future representation needs to support multiple joint conditions leading to one physical event is a design question raised by this example, not one the paper itself sets out to solve.

2.10 Comparison with Expert HAZOP

The paper compares the automatically generated results against the original expert HAZOP for the same tank. Table 2.2 reports the counts given in the paper.

Table 2.2: Counts of causes and consequences in the original expert HAZOP versus the automatically generated worksheet, as reported by Single et al.

	Original (expert)	**Automated**
Causes	22	33
Consequences	13	28

Constraint on interpreting this comparison

A larger count of generated causes and consequences must not be read as evidence that the automated system is more correct or more complete than the expert HAZOP. The paper itself emphasizes the continued need for expert qualitative assessment of the generated results; quantity is not a validity metric.

A further observation from the comparison is terminological rather than structural: the automated system and the human-authored HAZOP sometimes describe the same underlying physical situation using different terminology or a different level of abstraction. **Our interpretation:** this is directly relevant to any future text-extraction approach, since a system reading historical HAZOP report text would need to handle synonyms, varying abstraction levels, implicit causal relationships, and differing expert writing styles — the problem is not reducible to matching exact phrases, even in a setting (like this one) where the comparison is between two already-structured outputs rather than between structured output and free text.

2.11 Key Observations

The paper distinguishes conceptually between preventive and mitigative safeguards, and notes that safeguards can attach to different points along a causal chain rather than to the scenario as a whole (consistent with the causal-chain representation illustrated earlier in this paper). However, the authors are explicit that safeguard and recommendation selection depends heavily on company practices, regulations, technical standards, industry context, and risk assessment — meaning generated safeguards still require human expert evaluation before use. The overall workflow is therefore human-in-the-loop, not autonomous end to end: ontology → automated inference → generated scenarios → human expert evaluation → final assessment, with experts still needed for plausibility, likelihood, severity, safeguard adequacy, and recommendations.

2.12 Limitations and Dependencies

The following limitations and dependencies are identified from the paper; they are not exaggerated beyond what the source supports.

1. Causal knowledge is encoded into the ontology beforehand, not discovered at run time.
2. Result completeness depends directly on ontology completeness.
3. The demonstrated case is equipment- / process-unit-level, not plant-wide.
4. Plant-wide interactions between units are not fully addressed.
5. Process-unit and P&ID information is not automatically extracted in the demonstrated system — it is modeled manually.
6. Human expert validation of generated scenarios remains necessary.
7. Risk assessment (likelihood, severity) is not fully automated.
8. Safeguard selection remains context-dependent on company/regulatory practice.
9. New or unrepresented causal knowledge requires ontology extension by a knowledge engineer.
10. The demonstrated system is not a free-text HAZOP report understanding system.

The central dependency underlying most of the above can be summarized as a knowledge-acquisition bottleneck (Figure 2.5): the inference step is automated, but the acquisition and encoding of causal knowledge into the ontology is not.



Figure 2.5: Only the inference step (fourth box) is automated; the preceding knowledge-engineering step remains a manual, expert-driven dependency.

The paper's stated future work includes: integration with broader industrial/process ontologies, integration with ISO 15926, integration with OntoCAPE, incorporating safety and security considerations, a sharper distinction between safeguards and recommendations, extraction of process-unit/equipment information from intelligent P&IDs, and propagation of deviations/effects through upstream and downstream process nodes. **Our interpretation:** the "extraction of process-unit/equipment information from intelligent P&IDs" item is the closest the paper comes to acknowledging its own knowledge-acquisition bottleneck as future work, though it is framed around P&ID data rather than historical report text.

2.13 Relevance to Our Research

How this paper relates to our provisional direction — not a finalized gap

Single et al. demonstrate: structured HAZOP knowledge + a process-unit model + symbolic inference → multi-step HAZOP causal scenarios, including branching and multiple-cause chains. They do *not* demonstrate: raw HAZOP report text → automatic causal knowledge extraction → causal graph → reasoning. This distinction, first raised in the initial research framing, is now grounded in a full read of the paper rather than an abstract-level reading. It provides a reference architecture for the *reasoning* half of a possible pipeline — what structured causal representation and symbolic reasoning over it can look like — but says nothing about whether the *knowledge-acquisition* half (currently manual, per Section 2.8 and the limitations above) could be replaced by text-based extraction. This remains a research question, not a research gap: HAZOPCT and the Category C/D papers have not yet been read at this depth, and any one of them could already address this question in ways this paper does not.

2.14 Relevance of Single et al. to Our Research Direction

The detailed analysis above clarifies what Single et al. contributes to our project and, equally importantly, what remains outside its scope. Our current working hypothesis is not simply to extract entities from HAZOP reports. Given historical HAZOP reports, we want to investigate whether a system can recover the causal reasoning embedded in those reports and represent it in a form that a machine can subsequently reason over. The conceptual pipeline is:

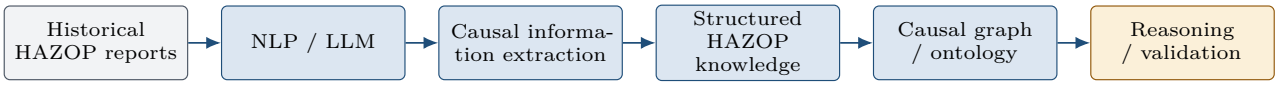


Figure 2.6: Conceptual pipeline currently guiding the project. The pipeline is a working hypothesis rather than a finalized system architecture or research gap.

This direction distinguishes information extraction from causal reasoning. Identifying a deviation, cause, consequence, and safeguard as isolated spans does not establish how those concepts are related. For example, a report may describe the chain *cooling-water failure → temperature increase → pressure increase → relief valve opens*. A basic NLP system may detect each concept correctly while losing the ordering, direction, intermediate events, or safeguard attachment. Our research interest is therefore in reconstructing the causal chain itself: information extraction is necessary, but information extraction is not causal reasoning.

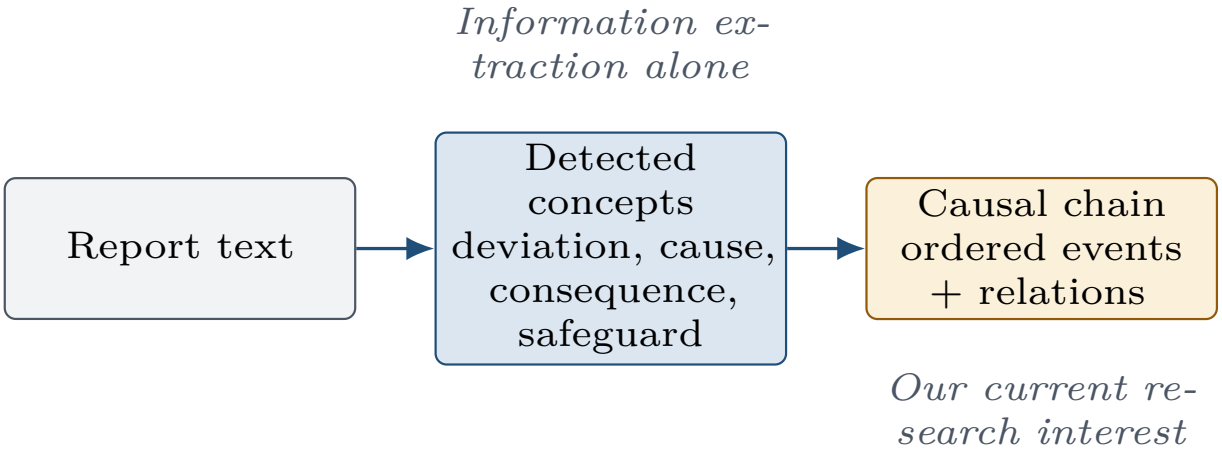


Figure 2.7: Detecting HAZOP concepts is an intermediate step; the intended research direction also requires recovery of the relations and ordering that form a causal chain.

Single et al. as a reasoning reference. Single et al. provides a useful reference for the structured-reasoning side of this project. Its ontology represents concepts including Process Unit, Equipment, Substance, Deviation, Cause, Effect, Super Cause, Consequence, and Safeguard, and its symbolic inference mechanism operates over relationships among those concepts. It consequently helps answer the question: *what could a machine-readable representation of HAZOP causal reasoning look like?* The ontology and its inferred chains provide a potential reasoning destination that an NLP/LLM pipeline may eventually need to recover from historical reports. This does not imply that the ontology is already the appropriate schema for our data; the literature study must determine which concepts and relations are required.

Difference between the two architectures. Conceptually, the architecture studied by Single et al. is:

Expert knowledge → ontology / structured process knowledge → symbolic inference → generated HAZOP scenarios.

Our current direction instead begins with historical documents:

Historical HAZOP reports → NLP / LLM → causal knowledge extraction → structured causal representation → reasoning / validation.

The important distinction is not that Single et al. fails to automate HAZOP analysis. It automates symbolic inference and demonstrates multi-step causal reasoning. Rather, its architecture primarily assumes that the relevant process and causal knowledge has already been formally represented. Our interest is in the earlier knowledge-recovery stage and in connecting that stage to reliable causal reasoning.

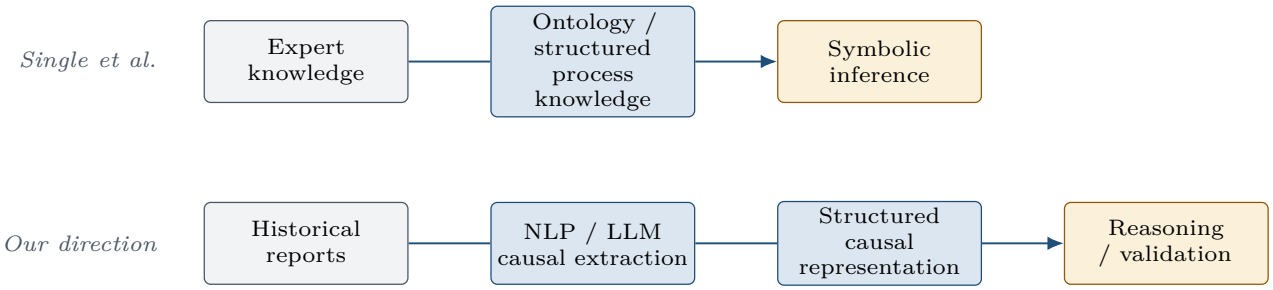


Figure 2.8: Single et al. assumes formally represented knowledge before inference, whereas our current direction investigates the knowledge-recovery stages preceding reasoning.

This distinction also separates the present research direction from ordinary document understanding or information-extraction work. Document understanding asks what information can be found in a document. The HAZOP problem we are formulating asks how the relationships and reasoning structure between pieces of information can be recovered. The object of interest is therefore not only the information present in the report, but the causal organization that makes that information useful for safety analysis.

Potential uses of a recovered causal structure. If a structured causal graph could be constructed, it could potentially support several future tasks. These are research questions, not capabilities provided by the current system:

- **Chain completeness:** given "cooling failure → temperature increases → ? → relief valve opens," determine whether a missing intermediate event can be proposed or flagged.
- **Causal validity:** distinguish a plausible sequence from a questionable one, such as "cooling failure → temperature decreases → pressure increases," using structured process-safety knowledge.
- **Semantic equivalence:** map statements such as "cooling water failure causes reactor overheating" and "loss of coolant results in elevated reactor temperature" to a common causal structure when their meanings are equivalent.
- **Missing-link inference:** propose relationships that are not explicitly recorded when historical reports contain only partial chains.

A possible representation is a graph in which nodes denote process-safety concepts or events and edges denote relations such as *causes*, *leads to*, *results in*, *prevented by*, or *mitigated by*:

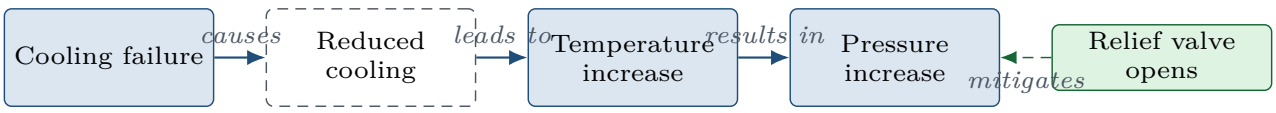


Figure 2.9: Illustrative causal graph. Nodes represent process-safety events and edges represent possible causal or safeguard relations; the schema is not yet finalized.

This is a conceptual example, not a finalized graph schema. A knowledge graph could provide the intermediate representation between report-level language and reasoning, but whether a knowledge graph, ontology, causal graph, or a hybrid is most appropriate requires further investigation.

Open questions for our investigation. Single et al. does not by itself answer how causal relationships should be extracted from raw HAZOP report text, how implicit causal relationships should be detected, or how different expert terminology should be normalized. It also leaves open how chains from multiple reports should

be merged, how previously unseen causal relationships should be handled, how LLM-generated relationships should be validated, and how uncertainty in extracted relationships should be represented. These are open questions that motivate the subsequent literature analysis; they are not yet being presented as the research gap.

The observations above define our current working research direction rather than a finalized research gap. Subsequent papers, particularly later knowledge-graph and HAZOP automation approaches, must be examined to determine which parts of this pipeline have already been addressed.

Takeaway. Single et al. is useful to our project not because it directly solves our intended problem, but because it provides an example of what structured HAZOP causal reasoning can look like. It gives us a potential reasoning destination: structured HAZOP knowledge → causal representation → symbolic reasoning. Our research is currently exploring how the earlier stages can be connected: historical HAZOP reports → NLP/LLM → causal structure → reasoning and validation. Later literature will determine whether this distinction is genuinely novel.

2.15 Questions for Subsequent Literature

The following questions are added to the literature comparison framework and can be applied to HAZOPCT and other remaining papers:

1. What is the input — raw HAZOP report, P&ID, structured process model, or ontology?
2. Does the method perform information extraction (from text or from another unstructured source)?
3. What representation is used — flat table, ontology, knowledge graph, or causal graph?
4. Does the method perform genuine causal reasoning, or only store labeled relationships?
5. Does it support multi-hop causal chains, as Single et al. do?
6. Does it infer missing causal relationships, or only reproduce encoded ones?
7. How is reasoning validity evaluated, and against what?
8. Is expert ground truth used for validation?
9. How dependent is the method on predefined/manually encoded knowledge?
10. Can it handle previously unseen causal relationships, or only combinations of known ones?
11. Is the system equipment-level, process-unit-level, or plant-wide in scope?

2.16 Conclusions

The full analysis of Single et al. (2020) shows that the paper performs genuine, multi-hop, branching causal reasoning over HAZOP concepts, evaluated against an expert worksheet on a real case study, and it does so entirely on the basis of manually encoded process and safety knowledge rather than on text extracted from historical reports. No research gap is being claimed on this basis. The open question for the project is whether the knowledge-acquisition step this system depends on — and does not attempt to automate — can be addressed by NLP/LLM-based extraction from historical report text, and whether the resulting extracted knowledge could be validated using reasoning structures similar to the one this paper demonstrates. HAZOPCT (Li & Zhao, 2025) is the next knowledge-graph-based alternative to examine before this question is pursued further.

Week 3 Literature Synthesis and Research Design

3.1 Objective

The objective of this week was to consolidate the literature studied during the previous weeks and use it to begin defining the research problem, case study, and experimental methodology for the project. Rather than treating the individual papers as isolated approaches, the analysis focuses on the different stages involved in automated HAZOP analysis: knowledge representation, information extraction, knowledge-graph construction, causal reasoning, missing-relationship inference, and direct HAZOP generation.

The purpose of this synthesis is not yet to claim a finalized research gap. Instead, it is to determine which parts of the HAZOP reasoning pipeline have already been addressed, what assumptions existing approaches make, and which questions should be investigated through a focused case study.

Week 3 at a glance

This week moves from reviewing individual automation approaches to defining the project's research design. The central working direction is to extract causal propositions from historical HAZOP reports, represent them explicitly, and evaluate whether the resulting structure supports multi-hop reasoning. The case study, schema, and ground-truth strategy remain provisional until candidate datasets have been examined.

3.2 From Automated HAZOP Generation to Structured Causal Reasoning

The literature studied so far suggests that "automating HAZOP" is not a single technical problem. Different approaches automate different stages of the overall process.

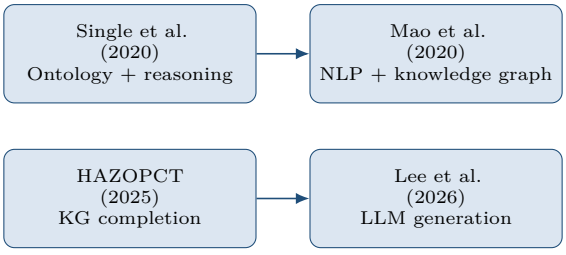
Single et al. (2020) demonstrate that HAZOP knowledge can be represented using an ontology and that symbolic inference can be used to generate multi-step HAZOP scenarios. Their system operates on an already structured process and safety representation, with expert knowledge encoded in the ontology before the reasoning process begins [12].

Mao et al. (2020) move the knowledge-acquisition problem closer to the document level by developing a process-safety knowledge graph from heterogeneous process-safety information. Their approach uses natural language processing for named-entity recognition and relation extraction from textual sources, including HAZOP reports. The resulting graph is then used for applications including information retrieval and causal-chain extraction [9].

Li and Zhao's HAZOPCT (2025) extends the knowledge-graph direction toward HAZOP completeness. Their approach constructs a HAZOP knowledge graph from multiple sources and applies graph-based learning to predict potentially missing relationships, allowing existing HAZOP information to be used for the identification of additional conclusions [8].

Lee et al. (2026) investigate a different direction: direct generation of complete HAZOP worksheets using multimodal large language models. Their experiments show that LLM-generated worksheets can achieve high textual similarity to expert-prepared worksheets, while the proportion of semantically valid generated scenarios remains considerably lower. This distinction is important because linguistic similarity alone does not establish the correctness of a causal or process-safety argument [6].

The four approaches therefore address different parts of the overall problem, as summarized in Figure 3.1.



Different stages of automated HAZOP analysis

Figure 3.1: The papers studied so far address different stages of automated HAZOP analysis: structured symbolic reasoning, knowledge extraction and graph construction, graph-based completion, and direct LLM-based generation.

3.3 A Layered View of the HAZOP Problem

The comparison indicates that several capabilities must be distinguished when discussing automated HAZOP analysis.

1. **Information extraction:** identifying relevant entities, events, and relationships from documents.
2. **Representation:** encoding extracted information using a structured representation such as a table, ontology, knowledge graph, or causal graph.
3. **Causal reasoning:** determining how deviations, causes, effects, consequences, and safeguards are related and how events propagate through intermediate states.
4. **Inference and validation:** identifying potentially missing relationships or scenarios and determining whether the resulting reasoning is consistent with process and safety knowledge.

These capabilities are related but are not interchangeable. In particular, identifying the concepts "cooling failure", "high temperature", and "pressure increase" does not by itself establish that they form an ordered causal sequence.

The distinction is illustrated in Figure 3.2.

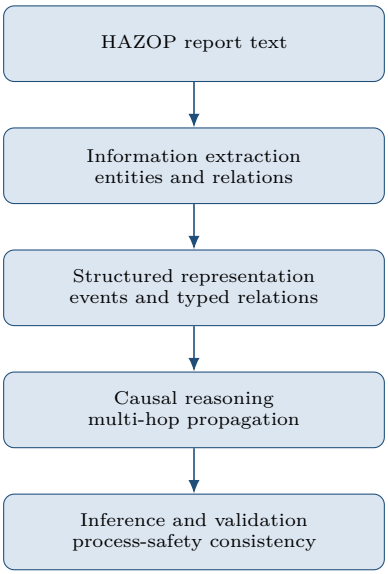


Figure 3.2: A layered view of the HAZOP problem. Information extraction is an intermediate step; causal reasoning additionally requires relationships, direction, ordering, and propagation between events.

This distinction is particularly relevant to the present project. A system may correctly identify individual HAZOP concepts while still failing to recover the reasoning connecting those concepts.

3.4 Why Causal Structure Is the Central Focus

The literature studied so far suggests that the causal structure of HAZOP information is more important than the presence of individual keywords alone.

Consider the following hypothetical description:

Cooling-water failure may increase reactor temperature, which can cause pressure buildup and ultimately result in loss of containment.

A conventional information-extraction system may identify:

cooling-water failure, *reactor temperature*, *pressure buildup*, *loss of containment*.

However, identifying these concepts does not establish the causal structure:

Cooling failure → Temperature increase → Pressure increase → Loss of containment.

The direction and ordering of these relationships are essential. A system that detects all four concepts but assigns incorrect relationships has not recovered the causal reasoning expressed by the original statement.

This distinction is also consistent with the progression of the literature. Single et al. demonstrate that HAZOP reasoning can contain multi-step and branching causal structures, while Mao et al. demonstrate how process-safety information can be organized into a knowledge graph [12, 9].

The present investigation therefore focuses on the connection between natural-language information and an explicit causal representation.

3.5 Current Working Research Pipeline

Based on the literature studied so far, the current working hypothesis is to investigate a pipeline that begins with historical HAZOP reports rather than with an already constructed process ontology.

The proposed high-level pipeline is:

Historical HAZOP reports → NLP/LLM extraction → causal information → structured causal representation → causal reasoning → validation.

Figure 3.3 presents the same idea in a compact system representation.

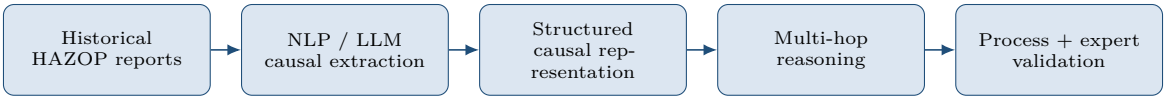


Figure 3.3: Current working pipeline for investigating structured HAZOP reasoning from historical reports. This is a working research hypothesis rather than a finalized system architecture.

The purpose of this pipeline is not to assume that a knowledge graph or a particular ontology is already the correct representation. Instead, the literature is being used to determine the representation and reasoning requirements that the case study must satisfy.

In particular, the following questions remain open:

1. Which HAZOP concepts need to be explicitly represented?
2. Which relation types are required to distinguish causality from simple association?
3. How should multi-step and branching causal chains be represented?
4. How should multiple causes leading to a common effect be represented?
5. How should safeguards be attached to causal chains?
6. How should uncertainty in extracted relationships be represented?
7. Can an extracted causal graph support reasoning over more than one causal hop?
8. Can missing or incomplete causal relationships be identified?

These questions will be used to define the case study and experimental methodology in the subsequent sections of this week's report.

3.6 Case-Study Design

The literature review established that the central objective of this research is not merely to identify HAZOP entities, but to recover the causal structure connecting deviations, causes, consequences, and safeguards. Therefore, the choice of case study is an important part of the experimental design.

Rather than selecting a process solely because a HAZOP dataset is available, the case study should provide sufficient causal structure for evaluating information extraction and subsequent reasoning. The selected process should also contain enough variation in terminology and causal descriptions to test whether the proposed NLP/LLM approach can generalize beyond a small collection of nearly identical sentences.

3.6.1 Requirements for the Case Study

The initial requirements for the case study are summarized in Figure 3.4.

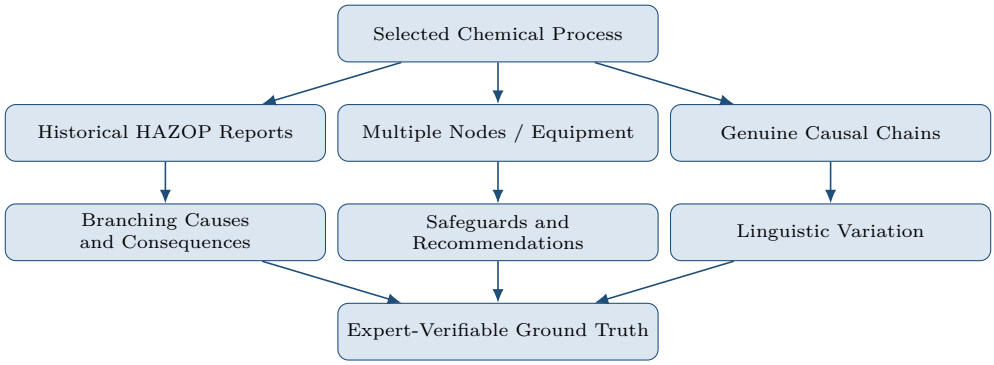


Figure 3.4: Initial requirements for selecting the case study.

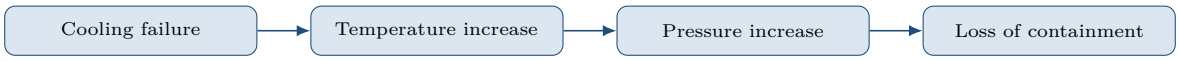
The following characteristics are considered desirable.

1. **Availability of HAZOP reports:** The case study should have access to sufficiently detailed HAZOP worksheets or reports containing deviations, causes, consequences, safeguards, and relevant process context.
2. **Multiple process nodes:** The process should contain multiple equipment items or HAZOP nodes so that the proposed method can be evaluated across different process contexts.
3. **Explicit causal relationships:** The reports should contain statements from which causal relationships can be recovered rather than only isolated labels or keywords.
4. **Multi-step causal chains:** The data should contain cases in which a cause produces an intermediate state or deviation that subsequently produces one or more consequences.
5. **Branching structure:** Ideally, individual deviations should have multiple possible causes and consequences. This is important because a simple one-to-one mapping would not adequately test causal graph construction.
6. **Safeguard information:** The HAZOP records should contain safeguards or recommendations so that the causal structure can eventually be connected to risk mitigation information.
7. **Linguistic variation:** Different HAZOP entries should express similar causal relationships using different terminology or sentence structures. This allows the extraction system to be tested for semantic rather than purely lexical matching.
8. **Expert-verifiable ground truth:** The extracted relationships and reconstructed causal chains should be capable of being checked against the original HAZOP analysis and, where possible, by a process-safety expert.

3.6.2 Why These Requirements Matter

The requirements are motivated by the distinction between information extraction and causal reasoning identified in the literature.

For example, consider a simplified process-safety description:



In this example, identifying the four concepts is only the first step. A useful system must also recover the relationships between them. The resulting structure can be represented as

$$C_1 \rightarrow D_1 \rightarrow E_1 \rightarrow C_2, \tag{3.1}$$

where the symbols represent successive causal states rather than independent HAZOP entities.

Consequently, a case study containing only isolated statements such as “high pressure”, “valve failure”, or “fire” would be insufficient for evaluating the main research objective. The dataset must contain enough contextual information to determine how such concepts are related.

3.6.3 HAZOP Entry as the Basic Unit of Analysis

The initial experimental unit will be an individual HAZOP entry or worksheet row. A conventional entry can be viewed as containing several semantically distinct components:



This representation should not be interpreted as claiming that every HAZOP entry forms a simple linear chain. Instead, it provides an initial decomposition of the information contained in a worksheet. The actual representation may contain multiple causes, intermediate states, several consequences, and safeguards attached to different parts of the causal structure.

3.6.4 From HAZOP Rows to Causal Propositions

A major objective of the case-study analysis will therefore be to transform the textual information in a HAZOP entry into explicit causal propositions.

For example, a textual entry could conceptually produce propositions of the form

$$\text{Cause} \xrightarrow{\text{causes}} \text{Deviation}, \tag{3.2}$$

$$\text{Deviation} \xrightarrow{\text{leads_to}} \text{Consequence}, \tag{3.3}$$

$$\text{Consequence} \xrightarrow{\text{mitigated_by}} \text{Safeguard}. \tag{3.4}$$

The precise relation vocabulary is intentionally left open at this stage. Determining whether these relations are sufficient, or whether additional relations are required to represent intermediate process states, is one of the questions to be investigated during the subsequent design phase.

3.6.5 Ground-Truth Construction

The original HAZOP reports will serve as the primary source of ground-truth information. However, the original worksheet structure may not directly provide all of the intermediate causal links required for evaluating multi-hop reasoning.

Therefore, the ground truth will be developed in stages:

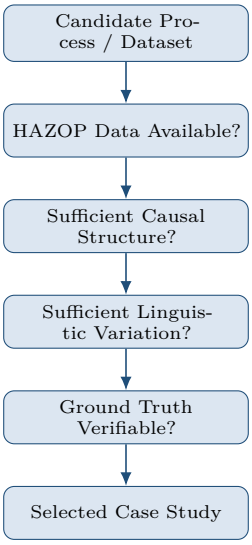
1. identify the relevant HAZOP entities in the source entry;
2. identify explicitly stated causal relationships;
3. identify multi-step relationships that are supported by the text and process context;
4. represent alternative causes and consequences where they exist;
5. associate safeguards with the corresponding causal scenario;
6. obtain expert review for relationships that require process interpretation.

This distinction is important because an automatically extracted graph should not be treated as ground truth merely because the model generated it. The reference structure must originate from the available HAZOP evidence and, where necessary, expert assessment.

3.6.6 Case-Study Selection Strategy

At this stage, a specific process will not be selected solely on the basis of convenience. Candidate datasets will instead be evaluated against the requirements defined above.

The selection can be formulated as a screening process:



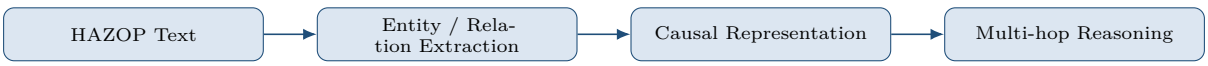
3.6.7 Current Status

The case-study process is therefore considered an open design decision at this point. The immediate objective is to identify a dataset that satisfies the causal and validation requirements rather than to optimize for dataset size alone.

This approach also allows the research design to remain independent of a particular process until the available HAZOP material has been examined. Once candidate reports are obtained, their structure, terminology, causal depth, branching, and available process context can be compared against the requirements established in this section.

3.6.8 Implications for the Experimental Design

The case study will eventually support evaluation at several levels:



This decomposition will make it possible to distinguish errors caused by incorrect extraction from errors introduced during graph construction or causal reasoning. Such separation is necessary for determining whether an observed failure is an NLP problem, a representation problem, or a reasoning problem.

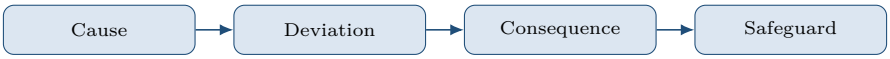
The next stage of the research will therefore focus on examining candidate HAZOP datasets and determining the exact causal schema that can be reliably annotated from them.

3.7 Causal Schema and Experimental Design

The previous sections established the literature landscape and the requirements for selecting a suitable HAZOP case study. The remaining task is to define how the extracted information will be represented, evaluated, and used for causal reasoning.

3.7.1 Initial Causal Representation

The working representation is based on typed entities and relationships rather than the original tabular HAZOP format. The initial structure is:



This should not be interpreted as a restriction to linear chains. A deviation may have multiple causes and consequences, while a consequence may itself become an intermediate state in a longer causal chain.

The representation can therefore be expressed as

$$G = (V, E),$$

where V contains entities and E contains typed relationships.

The initial entity types are:

Entity	Description
Process Node	HAZOP study location
Equipment	Physical equipment involved
Parameter	Process variable
Deviation	HAZOP deviation
Cause	Initiating or contributing condition
Consequence	Resulting outcome
Safeguard	Existing protective measure
Recommendation	Proposed protective action

The schema is provisional and will be revised after examination of the selected HAZOP data.

3.7.2 Causal Relations

The initial relation vocabulary is:

Relation	Meaning
causes	Cause produces a deviation or state
leads_to	One state produces another state or consequence
results_in	Consequence relation
mitigated_by	Protective measure associated with a scenario
associated_with	Contextual association
has_parameter	Equipment/node–parameter relationship
has_deviation	Node–deviation relationship

The final relation vocabulary will be determined from the actual language and causal structures present in the dataset.

3.7.3 Annotation Strategy

The basic annotation unit will be a causal proposition represented as

$$(h, r, t),$$

where h is the source entity, r is the relation, and t is the target entity.

For example,

$$(\text{Cooling Failure}, \text{causes}, \text{High Temperature})$$

followed by

$$(\text{High Temperature}, \text{leads_to}, \text{Pressure Increase}).$$

Multiple propositions can then be combined into a causal graph.

The annotation process will distinguish between:

1. entities and process context;

2. explicitly stated causal relationships;
3. multi-step relationships;
4. alternative causes and consequences;
5. safeguards and recommendations.

Implicit relationships that require substantial process interpretation will be marked separately rather than automatically treated as ground truth.

3.7.4 Proposed Research Pipeline

The complete working pipeline for the study is:



The extracted structure will then be compared against the annotated reference structure and evaluated for causal correctness.

3.7.5 Experimental Comparison

The initial experiments will compare different levels of extraction and reasoning rather than evaluating only a single model.

Experiment 1: Classical NLP baseline
A conventional pipeline using named entity recognition followed by relation extraction will provide a baseline for comparison.

Experiment 2: LLM structured extraction
An LLM will be prompted to extract HAZOP entities and typed causal relations into a predefined structured format.

Experiment 3: Explicit causal extraction
The extraction task will explicitly require identification of causal direction and intermediate states rather than simply identifying related entities.

Experiment 4: Causal graph reasoning
The extracted relations will be assembled into a graph and used for multi-hop reasoning.

Experiment 5: Missing-relation experiment
Known relationships can be temporarily removed from the graph and the system can be tested on whether it can recover the missing relationship or causal path.

The last experiment is particularly relevant to the distinction between extraction and reasoning. A system that merely reproduces information from the input is different from one that can use an existing causal structure to identify a missing relationship.

3.7.6 Evaluation Metrics

Evaluation will be performed at multiple levels.

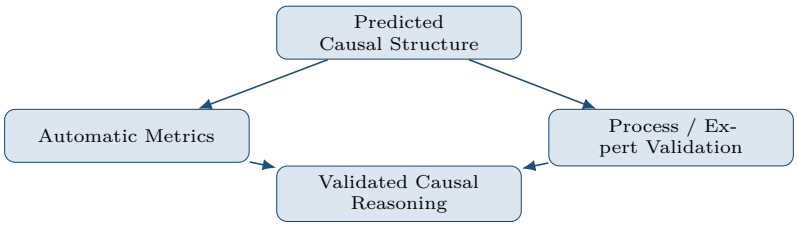
Level	Possible Metrics
Entity extraction	Precision, Recall, F1
Relation extraction	Precision, Recall, F1
Causal direction	Relation accuracy
Causal chains	Path Precision, Recall, F1
Multi-hop reasoning	1-hop / 2-hop / 3-hop accuracy
Missing relationships	Recovery accuracy, false positives
Safety validity	Expert/process-safety assessment

This separation is important because a system may achieve good entity recognition while still producing incorrect causal relationships.

Similarly, correct relation extraction does not necessarily imply successful multi-hop reasoning.

3.7.7 Validation Strategy

The final evaluation should therefore include both computational and domain-based validation.



Automatic metrics will measure agreement with the annotated reference structure, while expert validation will be necessary for relationships whose correctness depends on process knowledge or physical interpretation.

3.7.8 Working Research Question

Based on the literature examined so far, the working research question is:

> *Can natural-language processing and large language models reliably extract the causal structure embedded in HAZOP reports and represent it in a form that supports explicit multi-hop causal reasoning?*

This remains a working research question rather than a finalized claim of novelty.

The investigation will focus on three supporting questions:

1. Can NLP/LLMs reliably extract causal relationships from HAZOP text?
2. Does an explicit causal representation improve multi-hop reasoning compared with direct generation of HAZOP conclusions?
3. Can causal reasoning identify missing or overlooked relationships while maintaining process-safety validity?

3.7.9 Expected Research Direction

The current understanding of the project can therefore be summarized as:

$$\boxed{\begin{array}{c}\text{Historical HAZOP Reports} \rightarrow \text{Causal Extraction} \rightarrow \\ \text{Structured Representation} \rightarrow \text{Multi-hop Reasoning} \rightarrow \text{Validation}\end{array}}$$

The specific representation—knowledge graph, ontology, causal graph, or a hybrid approach—will not be fixed until the case-study data and annotation requirements have been examined.

This preserves flexibility while keeping the central objective unchanged: recovering and reasoning over the causal structure embedded in HAZOP documentation.

3.7.10 Next Steps

The immediate next steps are:

1. identify candidate HAZOP datasets;

2. examine their structure and availability;

3. select the case study using the requirements defined earlier;

4. inspect representative HAZOP entries;

5. refine the entity and relation schema;

6. define the annotation format;

7. construct an initial annotated subset;

8. establish the first extraction baseline.

These steps will provide the transition from literature and research design into the implementation and experimental phase of the project.

Bibliography

[1] Abdul Aziz, Salim Ahmed, and Faisal I. Khan. An ontology-based methodology for hazard identification and causation analysis. *Process Safety and Environmental Protection*, 123:87–98, 2019.

[2] Ali Ekramipooya, Mehrdad Boroushaki, and Davood Rashtchian. Application of natural language processing and machine learning in prediction of deviations in the HAZOP study worksheet: A comparison of classifiers. *Process Safety and Environmental Protection*, 176:65–73, 2023.

[3] Ali Ekramipooya, Mehrdad Boroushaki, and Davood Rashtchian. Predicting possible recommendations related to causes and consequences in the HAZOP study worksheet using natural language processing and machine learning: BERT, clustering, and classification. *Journal of Loss Prevention in the Process Industries*, 89:105310, 2024.

[4] Xiayuan Feng, Yiyang Dai, Xu Ji, and Li Zhou. HAZOP causal knowledge description model based on ontology. *Chemical Industry and Engineering Progress*, 40(11):6044–6053, 2021. Original article in Chinese.

[5] Xiayuan Feng, Yiyang Dai, Xu Ji, Li Zhou, and Yagu Dang. Application of natural language processing in HAZOP reports. *Process Safety and Environmental Protection*, 155:41–48, 2021.

[6] Junseo Lee, Sunhwa Park, Sehyeon Oh, and Byungchol Ma. Can large language models automate the HAZOP process without human intervention? *Safety Science*, 194:107039, 2026.

[7] Zhiyuan Li and Jinsong Zhao. Constructing a knowledge graph for automated HAZOP analysis. In *Computer Aided Chemical Engineering*, volume 53, pages 1291–1296. Elsevier, 2024.

[8] Zhiyuan Li and Jinsong Zhao. HAZOPCT: A HAZOP analysis completeness tool based on knowledge graph reasoning. *Process Safety and Environmental Protection*, 197:107025, 2025.

[9] Shuai Mao, Yunmeng Zhao, Jinhe Chen, Bing Wang, and Yang Tang. Development of process safety knowledge graph: A case study on delayed coking process. *Computers and Chemical Engineering*, 143:107094, 2020.

[10] Shibly Rahman, Faisal Khan, Brian Veitch, and Paul Amyotte. ExpHAZOP+: Knowledge-based expert system to conduct automated HAZOP analysis. *Journal of Loss Prevention in the Process Industries*, 22(4):373–380, 2009.

[11] Manuel Rodríguez and José Luis de la Mata. Automating HAZOP studies using d-higraphs. *Computers and Chemical Engineering*, 45:102–113, 2012.

[12] Johannes I. Single, Jürgen Schmidt, and Jens Denecke. Ontology-based computer aid for the automation of HAZOP studies. *Journal of Loss Prevention in the Process Industries*, 68:104321, 2020.

[13] J. R. Taylor. Automated HAZOP revisited. *Process Safety and Environmental Protection*, 111:635–651, 2017.

[14] Ramesh Vaidhyanathan and Venkat Venkatasubramanian. Digraph-based models for automated HAZOP analysis. *Reliability Engineering and System Safety*, 50(1):33–49, 1995.

[15] Ramesh Vaidhyanathan and Venkat Venkatasubramanian. Experience with an expert system for automated HAZOP analysis. *Computers and Chemical Engineering*, 20(Supplement 2):S1589–S1594, 1996.

[16] Feng Wang and Wunan Gu. Intelligent HAZOP analysis method based on data mining. *Journal of Loss Prevention in the Process Industries*, 80:104911, 2022.

[17] Zhenhua Wang, Beike Zhang, and Dong Gao. Text mining of hazard and operability analysis reports based on active learning. *Processes*, 9(7):1178, 2021.

[18] Zhenhua Wang, Beike Zhang, and Dong Gao. A novel knowledge graph development for industry design: A case study on indirect coal liquefaction process. *Computers in Industry*, 139:103647, 2022.